

The Missing Frobenius

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Abstract

Over a finite field $\mathbb{F}_q$, there is a canonical symmetry: the Frobenius $x \mapsto x^q$. It is the reason the Weil conjectures are provable. Over $\mathbb{Q}$, there is no global Frobenius. This is the reason the Riemann Hypothesis is not provable by the same method. The missing Frobenius is not a conjecture. It is an absence. What is absent, and why, and what its presence would mean.

1 What the Frobenius Is

Over the finite field $\mathbb{F}_q$ with $q = p^n$ elements, the Frobenius morphism is:

$$\varphi_q : x \longmapsto x^q.$$

It fixes exactly the elements of $\mathbb{F}_q$ and permutes everything else. It is a canonical element of $\text{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q)$: the generator, the clock of the field.

For a variety $X/\mathbb{F}_q$, the Frobenius acts on the étale cohomology $H_{\text{ét}}^*(X_{\overline{\mathbb{F}_q}}, \mathbb{Q}_\ell)$. Its eigenvalues are the arithmetic of X : the number of points over $\mathbb{F}_{q^n}$ is determined by the trace of φ_q^n on cohomology.

This is the Lefschetz trace formula, and it is exact. The Frobenius encodes everything.

Theorem 1 (Weil, Deligne [1, 2]). *For a smooth projective variety $X/\mathbb{F}_q$, the eigenvalues of φ_q on $H_{\text{ét}}^i(X, \mathbb{Q}_\ell)$ are algebraic integers of absolute value $q^{i/2}$. In particular, the zeros of the zeta function of X lie on the lines $\text{Re}(s) = i/2$ for $i = 0, 1, \dots, 2 \dim X$.*

This is the Riemann Hypothesis over finite fields. It is proved. It is proved because the Frobenius exists.

2 Why There Is No Global Frobenius

Over $\mathbb{Q}$, there is no analogue.

At each prime p , there is a local Frobenius: $\varphi_p \in \text{Gal}(\bar{\mathbb{Q}}_p/\mathbb{Q}_p)$, well-defined up to conjugacy. The local Frobenii are the building blocks of the Galois group $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$. But there is no single element of the global Galois group that acts as the Frobenius everywhere simultaneously.

The reason is structural. Over $\mathbb{F}_q$, the Galois group is $\hat{\mathbb{Z}}$, generated by one element (φ_q). Over $\mathbb{Q}$, the Galois group $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ is profinite, non-abelian, and vastly more complex. It has no preferred generator. It has local Frobenii at each prime, but they do not cohere into a global one.

Remark 2. This is not a theorem that says “the global Frobenius does not exist.” It is an observation about the structure of $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$: it has no element with the properties the Frobenius would need to have. The absence is structural, not accidental.

3 What the Frobenius Would Do

If a global Frobenius $\varphi_{\mathbb{Q}}$ existed, acting on the motivic cohomology $H_{\text{mot}}^*(\text{Spec } \mathbb{Q})$ with the right eigenvalue properties, the following would follow:

1. **Riemann Hypothesis.** The zeros of $\zeta(s)$ would be the eigenvalues of $\varphi_{\mathbb{Q}}$

on $H_{\text{mot}}^1(\text{Spec } \mathbb{Z})$. By the Weil-type bound, they would have absolute value $q^{1/2}$ for the appropriate q — placing them on $\text{Re}(s) = 1/2$.

2. **Birch–Swinnerton-Dyer.** The rank of $E(\mathbb{Q})$ would be the multiplicity of the eigenvalue 1 of $\varphi_{\mathbb{Q}}$ on $H_{\text{mot}}^1(E)$. The order of vanishing of $L(E, s)$ at $s = 1$ would be this multiplicity.
3. **Hodge conjecture (over $\mathbb{Q}$).** The global Frobenius would distinguish algebraic classes from transcendental ones: an algebraic cycle has Frobenius eigenvalue a root of unity; a transcendental class does not. The (p, p) condition would follow from Frobenius acting as multiplication by q^p .

Three walls. One missing symmetry.

4 Candidates for the Missing Frobenius

The absence has been noticed. Several structures have been proposed as its replacement or incarnation.

The Fargues–Fontaine curve. Fargues and Fontaine constructed a curve X_{FF} such that p -adic representations of $\text{Gal}(\bar{\mathbb{Q}}_p/\mathbb{Q}_p)$ correspond to vector bundles on X_{FF} . The Frobenius on X_{FF} is the p -adic Frobenius. This works locally. The global version is not yet constructed.

The Λ -ring structure. The ring $\mathbb{Z}$ carries a Λ -ring structure (Adams operations ψ^p for each prime p) which behaves like a family of Frobenii. This is the basis of the theory of Λ -schemes and related to the “field with one element” $\mathbb{F}_1$. A geometry over $\mathbb{F}_1$ would have a single Frobenius reducing to φ_p at each prime.

Connes’ scaling action. In the Bost–Connes system [3], the time evolution $\sigma_t(f) = f \cdot |\cdot|^{it}$ plays the role of the Frobenius. The KMS states at inverse temperature $\beta > 1$ are indexed by $\hat{\mathbb{Z}}^* \cong \text{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$. At $\beta = 1$ (the phase transition), the symmetry breaks. This is not quite a Frobenius, but it is the closest operator-algebraic analogue.

The absolute Galois group. The deepest version: $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ itself might act on the motivic cohomology of $\text{Spec } \mathbb{Z}$ in a way that recovers the Frobenius at each prime. This is what the Langlands program would give, if the global Langlands correspondence were fully established. The global Frobenius would not be a single element but a representation of the whole group.

5 The Grammar of the Absence

The grammar paper [4] identifies the local-to-global passage as the wall. The missing Frobenius is the reason for the wall.

Over $\mathbb{F}_q$: the local and global Frobenii coincide. The grammar is complete. The Riemann Hypothesis is provable.

Over $\mathbb{Q}$: the local Frobenii exist but do not cohere globally. The grammar is incomplete. The Riemann Hypothesis is not provable by the same method.

What is needed is not a proof within the current grammar. What is needed is a global Frobenius — or something that does what the global Frobenius would do: a canonical symmetry of the arithmetic of $\mathbb{Q}$ that acts on the motivic cohomology with the right eigenvalue properties.

When that symmetry is found — if it is found — the proof will be the coboundary it generates. The wall will not be overcome. It will dissolve. The grammar will have been extended to include what was absent.

References

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